Appendix D

Glossary

A

ABM Asynchronous balanced mode

ACE Association control element

ACE Asynchronous communications element. Similar to UART

ACK Acknowledge (ASCII - control F)

Active filter Active circuit devices (usually amplifiers), with passive circuit elements (resistors and capacitors) and which have characteristics that more closely match ideal filters than do passive filters.

Active passive device Device capable of supplying the current for the loop (active) or one that must draw its power from connected equipment (passive).

ADCCP Advanced data communication control procedure

ADDR Address field

Address A normally unique designator for location of data or the identity of a peripheral device that allows each device on a single communications line to respond to its own message.

Algorithm Normally used as a basis for writing a computer program. This is a set of rules with a finite number of steps for solving a problem.

Alias frequency A false lower frequency component that appears in data reconstructed from original data acquired at an insufficient sampling rate (which is less than two (2) times the maximum frequency of the original data).

ALU Arithmetic logic unit

Amplitude flatness A measure of how close to constant the gain of a circuit remains over a range of frequencies.

Amplitude modulation A modulation technique (also referred to as AM or ASK) used to allow data to be transmitted across an analog network, such as a switched telephone network. The amplitude of a single (carrier) frequency is varied or modulated between two levels – one for binary 0 and one for binary 1.

Analog A continuous real time phenomena where the information values are represented in a variable and continuous waveform.

ANSI American National Standards Institute – the principal standards development body in the USA.

APM Alternating pulse modulation

Appletalk A proprietary computer networking standard initiated by the Apple Computer for use in connecting the Macintosh range of computers and peripherals (including laser writer printers). This standard operates at 230 kbps.

Application layer The highest layer of the seven layer ISO/OSI reference model structure, which contains all user or application programs.

Arithmetic logic unit The element(s) in a processing system that perform(s) the mathematical functions such as addition, subtraction, multiplication, division, inversion, AND, OR, NAND and NOR.

ARP Address resolution protocol
A transmission control protocol/ Internet protocol (TCP/IP) process that maps an IP address to Ethernet address, required by TCP/IP for use with Ethernet.

ARQ Automatic request for transmission
A request by the receiver for the transmitter to retransmit a block or frame because of errors detected in the originally received message.

AS Australian standard

ASCII American standard code for information interchange. A universal standard for encoding alphanumeric characters into 7 or 8 binary bits. Drawn up by ANSI to ensure compatibility between different computer systems.

AS-i Actuator sensor interface

ASIC Application specific integrated circuit

ASK Amplitude shift keying – see Amplitude modulation

ASN.1 Abstract syntax notation 1 – an abstract syntax used to define the structure of the protocol data units associated with a particular protocol entity.

Asynchronous Communications where characters can be transmitted at an arbitrary unsynchronized point in time and where the time intervals between transmitted characters may be of varying lengths. Communication is controlled by start and stop bits at the beginning and end of each character.

Attenuation The decrease in the magnitude of strength (or power) of a signal. In cables, generally expressed in dB per unit length.

AWG American wire gauge

B

Balanced circuit A circuit so arranged that the impressed voltages on each conductor of the pair are equal in magnitude but opposite in polarity with respect to ground.

Bandpass filter A filter that allows only a fixed range of frequencies to pass through. All other frequencies outside this range (or band) are sharply reduced in magnitude.

Bandwidth The range of frequencies available expressed as the difference between the highest and lowest frequencies is expressed in Hertz (or cycles per second).

Base address A memory address that serves as the reference point. All other points are located by offsetting in relation to the base address.

Baseband Baseband operation is the direct transmission of data over a transmission medium without the prior modulation on a high frequency carrier band.

Baud Unit of signaling speed derived from the number of events per second (normally bits per second). However if each event has more than one bit associated with it the baud rate and bits per second are not equal.

Baudot Data transmission code in which five bits represent one character. 64 alphanumeric characters can be represented. This code is used in many teleprinter systems with one start bit and 1.42 stop bits added.

BCC Block check calculation

BCC Block check character – error checking scheme with one check character; a good example being block sum check.

BCD Binary coded decimal
A code used for representing decimal digits in a binary code.

BEL Bell (ASCII for control-G)

Bell 212 An AT&T specification of full duplex, asynchronous or synchronous 1200 baud data transmission for use on the public telephone networks.

BER Bit error rate

BERT/BLERT Bit error rate/block error rate testing – an error checking technique that compares a received data pattern with a known transmitted data pattern to determine transmission line quality.

BIN Binary digits

BIOS Basic input/output system

Bipolar A signal range that includes both positive and negative values.

BISYNC Binary synchronous communications protocol

Bit Derived from "**BI**nary Digi**T**", a one or zero condition in the binary system.

Bit stuffing with zero bit insertion A technique used to allow pure binary data to be transmitted on a synchronous transmission line. Each message block (frame) is encapsulated between two flags that are special bit sequences. Then if the message data contains a possibly similar sequence, an additional (zero) bit is inserted into the data stream by the sender, and is subsequently removed by the receiving device. The transmission method is then said to be data transparent.

Bits per second (bps) Unit of data transmission rate.

Block sum check This is used for the detection of errors when data is being transmitted. It comprises a set of binary digits (bits) which are the modulo 2 sum of the individual characters or octets in a frame (block) or message.

Bridge A device to connect similar subnetworks without its own network address. Used mostly to reduce the network load.

Broadband A communications channel that has greater bandwidth than a voice grade line and is potentially capable of greater transmission rates. Opposite of baseband. In wideband operation the data to be transmitted are first modulated on a high frequency carrier signal. They can then be simultaneously transmitted with other data modulated on a different carrier signal on the same transmission medium.

Broadcast A message on a bus intended for all devices that requires no reply.

BS Backspace (ASCII Control-H)

BS British standard

BSC Bisynchronous transmission
A byte or character oriented communication protocol that has become the industry standard (created by IBM). It uses a defined set of control characters for synchronized transmission of binary coded data between stations in a data communications system.

BSP Binary synchronous protocol

Buffer An intermediate temporary storage device used to compensate for a difference in data rate and data flow between two devices (also called a spooler for interfacing a computer and a printer).

Burst mode A high-speed data transfer in which the address of the data is sent followed by back to back data words while a physical signal is asserted.

Bus A data path shared by many devices with one or more conductors for transmitting signals, data or power.

Byte A term referring to eight associated bits of information; sometimes called a 'character'.

C

CAN Controller area network

Capacitance Storage of electrically separated charges between two plates having different potentials. The value is proportional to the surface area of the plates and inversely proportional to the distance between them.

Capacitance (mutual) The capacitance between two conductors with all other conductors, including shield, short-circuited to the ground.

CATV Community Antenna Television

CCITT See ITU

Cellular polyethylene Expanded or 'foam' polyethylene consisting of individual closed cells suspended in a polyethylene medium.

Character Letter, numeral, punctuation, control figure or any other symbol contained in a message.

Characteristic impedance The impedance that, when connected to the output terminals of a transmission line of any length, makes the line appear infinitely long. The ratio of voltage to current at every point along a transmission line on which there are no standing waves.

CIC Controller in charge

Clock The source(s) of timing signals for sequencing electronic events e.g. synchronous data transfer.

CMD Command byte

CMR Common mode rejection

CMRR Common mode rejection ratio

CMV Common mode voltage

Common carrier A private data communications utility company that furnishes communications services to the general public.

Composite link The line or circuit connecting a pair of multiplexers or concentrators; the circuit carrying multiplexed data.

Contention The facility provided by the dial network or a data PABX which allows multiple terminals to compete on a first come, first served basis for a smaller number of computer posts.

CPU Central processing unit

CR Carriage return (ASCII control-M)

CRC Cyclic redundancy check – an error-checking mechanism using a polynomial algorithm based on the content of a message frame at the transmitter and included in a field appended to the frame. At the receiver, it is then compared with the result of the calculation that is performed by the receiver. Also referred to as CRC-16.

CRL Communication relationship list

Cross talk A situation where a signal from a communications channel interferes with an associated channel's signals.

Crossed planning Wiring configuration that allows two DTE or DCE devices to communicate. Essentially it involves connecting pin 2 to pin 3 of the two devices.

Crossover In communications, a conductor that runs through the cable and connects to a different pin number at each end.

CSMA/CD Carrier sense multiple access/collision detection – when two senders transmit at the same time on a local area network; they both cease transmission and signal that a collision has occurred. Each then tries again after waiting for a predetermined time period.

CTS Clear to send

Current Loop Communication method that allows data to be transmitted over a longer distance with a higher noise immunity level than with the standard EIS-232-C voltage method. A mark (a binary 1) is represented by current of 20 mA and a space (or binary 0) is represented by the absence of current.

D

DAQ Data acquisition

Data integrity A performance measure based on the rate of undetected errors.

Data link layer This corresponds to layer 2 of the ISO reference model for open systems interconnection. It is concerned with the reliable transfer of data (no residual transmission errors) across the data link being used.

Data reduction The process of analyzing large quantities of data in order to extract some statistical summary of the underlying parameters.

Datagram A type of service offered on a packet-switched data network. A datagram is a self contained packet of information that is sent through the network with minimum protocol overheads.

DCD Data carrier detect

DCE Data communications equipment or data circuit-terminating equipment – devices that provide the functions required to establish, maintain, and terminate a data transmission connection. Normally it refers to a modem.

DCS Distributed control systems

Decibel (dB) A logarithmic measure of the ratio of two signal levels where $dB = 20\log_{10} V1/V2$ or where $dB = 10\log_{10} P1/P2$ and where V refers to voltage or P refers to power. Note that it has no units of measurement.

Default A value or setup condition assigned, which is automatically assumed for the system unless otherwise explicitly specified.

Delay distortion Distortion of a signal caused by the frequency components making up the signal having different propagation velocities across a transmission medium.

DES Data encryption standard

DFM Direct frequency modulation

Dielectric constant (E) The ratio of the capacitance using the material in question as the dielectric, to the capacitance resulting when the material is replaced by air.

Digital A signal which has definite states (normally two).

DIN Deutsches Institut Fur Normierung

DIP Dual in line package, referring to integrated circuits and switches.

Direct memory access A technique of transferring data between the computer memory and a device on the computer bus without the intervention of the microprocessor. Also abbreviated to DMA.

DISC Disconnect

DLE Data link escape (ASCII character)

DNA Distributed network architecture

DPI Dots per inch

DPLL Digital phase locked loop

DR Dynamic range
The ratio of the full-scale range (FSR) of a data converter to the smallest difference it can resolve. DR = 2n where n is the resolution in bits.

Driver software A program that acts as the interface between a higher-level coding structure and the lower level hardware/firmware component of a computer.

DSP Digital signal processing

DSR Data set ready or DCE ready in EIA-232D/E – A EIA-232 modem interface control signal which indicates that the terminal is ready for transmission.

DTE Data terminal equipment – devices acting as data source or data sink, or both.

DTR Data terminal ready or DTE ready in EIA-232D/E

Duplex The ability to send and receive data simultaneously over the same communications line.

E

EBCDIC Extended binary coded decimal interchange code
An eight bit character code used primarily in IBM equipment. The code allows for 256 different bit patterns.

EDAC Error detection and correction

EFTPOS Electronic funds transfer at the point of sale

EIA Electronic Industries Association – a standards organization in the USA specializing in the electrical and functional characteristics of interface equipment.

EISA Enhanced industry standard architecture

EMI/RFI Electromagnetic interference/radio frequency interference 'background noise' that could modify or destroy data transmission.

EMS Expanded memory specification

Emulation The imitation of a computer system performed by a combination of hardware and software that allows programs to run between incompatible systems.

ENQ Enquiry (ASCII Control-E)

EOT End of transmission (ASCII Control-D)

EPA Enhanced performance architecture

EPR Earth potential rise

EPROM Erasable programmable read only memory – non-volatile semiconductor memory that is erasable in an ultra violet light and reprogrammable.

Error rate The ratio of the average number of bits that will be corrupted to the total number of bits that are transmitted for a data link or system.

ESC Escape (ASCII character)

ESD Electrostatic discharge

ETB End of transmission block

Ethernet Name of a widely used LAN, based on the CSMA/CD bus access method (IEEE 802.3). Ethernet is the basis of the TOP bus topology.

ETX End of text (ASCII control-C)

Even parity A data verification method normally implemented in hardware in which each character must have an even number of 'ON' bits.

F

Farad Unit of capacitance whereby a charge of one coulomb produces a one volt potential difference.

FAS Fieldbus access sublayer

FCC Federal communications commission

FCS Frame check sequence – a general term given to the additional bits appended to a transmitted frame or message by the source to enable the receiver to detect possible transmission errors.

FDM Frequency division multiplexer – a device that divides the available transmission frequency range in narrower bands, each of which is used for a separate channel.

FIB Factory information bus

FIFO First in, first out

Filled cable A telephone cable construction in which the cable core is filled with a material that will prevent moisture from entering or passing along the cable.

FIP Factory instrumentation protocol

Firmware A computer program or software stored permanently in PROM or ROM or semi-permanently in EPROM.

Flame retardancy The ability of a material not to propagate flame once the flame source is removed.

Flow control The procedure for regulating the flow of data between two devices preventing the loss of data once a device's buffer has reached its capacity.

FMS Fieldbus message specification

FNC Function byte

Frame The unit of information transferred across a data link. Typically, there are control frames for link management and information frames for the transfer of message data.

Frequency modulation A modulation technique (abbreviated to FM) used to allow data to be transmitted across an analog network where the frequency is varied between two levels – one for binary '0'

	and one for binary '1'. Also known as frequency shift keying (or FSK).
Frequency	Refers to the number of cycles per second.
FRMR	Frame reject
FSK	Frequency shift keying, see frequency modulation
Full duplex	Simultaneous two-way independent transmission in both directions (4 wire). See Duplex.

G

G	Giga (metric system prefix – 10^9)
Gateway	A device to connect two different networks which translates the different protocols.
GMSK	Gaussian minimum shift keying
GPIB	General purpose interface bus – an interface standard used for parallel data communication, usually used for controlling electronic instruments from a computer. Also known as IEEE 488 standard.
Ground	An electrically neutral circuit that has the same potential as the earth. A reference point for an electrical system also intended for safety purposes.

H

Half duplex	Transmissions in either direction, but not simultaneously.
Hamming Distance	A measure of the effectiveness of error checking. The higher the Hamming distance (HD) index, the safer is the data transmission.
Handshaking	Exchange of predetermined signals between two devices establishing a connection.
Hardware	Refers to the physical components of a device, such as a computer, sensor, controller or data communications system. These are the physical items that one can see.
HART	Highway addressable remote transducers
HDLC	High level data link control The international standard communication protocol defined by ISO to control the exchange of data across either a point-to-point data link or a multidrop data link.
Hertz (Hz)	A term replacing cycles per second as a unit of frequency.
Hex	Hexadecimal
HF	High frequency
Host	This is normally a computer belonging to a user that contains (hosts) the communication hardware and software necessary to connect the computer to a data communications network.

HSE High speed Ethernet

I

I/O address A method that allows the CPU to distinguish between different boards in a system. All boards must have different addresses.

IA5 International alphabet number 5

IC Integrated circuit

ICS Instrumentation and control system

IDF Intermediate distribution frame

IEC International Electrotechnical Commission

IEE Institution of Electrical Engineers – an American based international professional society that issues its own standards and is a member of ANSI and ISO.

IEEE Institute of Electrical and Electronic Engineers

IFC International Fieldbus Consortium

ILD Injection laser diode

Impedance The total opposition that a circuit offers to the flow of alternating current or any other varying current at a particular frequency. It is a combination of resistance R and reactance X, measured in Ohms.

Inductance The property of a circuit or circuit element that opposes a change in current flow, thus causing current changes to lag behind voltage changes. It is measured in henrys.

Insulation resistance (IR) That resistance offered by insulation to an impressed dc voltage, tending to produce a leakage current though the insulation.

Interface A shared boundary defined by common physical interconnection characteristics, signal characteristics and measurement of interchanged signals.

Interrupt handler The section of the program that performs the necessary operation to service an interrupt when it occurs.

Interrupt An external event indicating that the CPU should suspend its current task to service a designated activity.

IP Internet protocol

IRQ Interrupt request line

ISA Industry Standard Architecture (for IBM Personal Computers)

ISB Intrinsically safe barrier

ISDN Integrated services digital network – the new generation of worldwide telecommunications network that utilizes digital

techniques for both transmission and switching. It supports both voice and data communications.

ISO International Standards Organization

ISP Interoperable systems project

ISR Interrupt service routine, see interrupt handler

ITB End of intermediate block

ITS Interface terminal strip

ITU International Telecommunications Union – formerly CCITT (Consultative Committee International Telegraph and Telephone). An international association that sets worldwide standards (e.g. V.21, V.22, V.22bis).

J

Jumper A wire connecting one or more pins on the one end of a cable only.

K

k (kilo) This is 2^{10} or 1024 in computer terminology,
e.g. 1 kB = 1024 bytes.

L

LAN Local area network – a data communications system confined to a limited geographic area typically about 10 km with moderate to high data rates (100 kbps to 50 Mbps). Some type of switching technology is used, but common carrier circuits are not used.

LAN Local area network. See Local area network.

LAP-M Link access protocol modem

LAS Link active scheduler

LCD Liquid crystal display – a low-power display system used on many laptops and other digital equipment.

LDM Limited distance modem – a signal converter which conditions and boosts a digital signal so that it may be transmitted further than a standard EIA-232 signal.

Leased (or Private) line A private telephone line without inter-exchange switching arrangements.

LED Light emitting diode
A semiconductor light source that emits visible light or infra red radiation.

LF Line feed (ASCII Control-J)

Line driver A signal converter that conditions a signal to ensure reliable transmission over an extended distance.

Line turnaround The reversing of transmission direction from transmitter to receiver or vice versa when a half duplex circuit is used.

Linearity A relationship where the output is directly proportional to the input.

Link layer Layer 2 of the ISO/OSI reference model. Also known as the data link layer.

Listener A device on the GPIB bus that receives information from the bus.

LLC Logical link control (IEEE 802)

LLI Lower layer interface

Loaded line A telephone line equipped with loading coils to add inductance in order to minimize amplitude distortion.

Loop resistance The measured resistance of two conductors forming a circuit.

Loopback Type of diagnostic test in which the transmitted signal is returned on the sending device after passing through all, or a portion of, a data communication link or network. A loopback test permits the comparison of a returned signal with the transmitted signal.

LRC Longitudinal redundancy check

LSB Least significant bits – the digits on the right hand side of the written HEX or BIN codes.

LSD Least significant digit

M

M Mega. Metric system prefix for 10^6.

m Meter. Metric system unit for length.

MAC Media Access Control (IEEE 802).

MAN Metropolitan Area Network

Manchester encoding Digital technique (specified for the IEEE 802.3 Ethernet baseband network standard) in which each bit period is divided into two complementary halves; a negative to positive voltage transition in the middle of the bit period designates a binary '1', whilst a positive to negative transition represents a '0'. The encoding technique also allows the receiving device to recover the transmitted clock from the incoming data stream (self clocking).

MAP 3.0 Standard profile for manufacturing developed by MAP.

MAP Manufacturing automation protocol – a suite of network protocols originated by General Motors, which follow the seven layers of the OSI model. A reduced implementation is referred to as a mini-MAP.

Mark This is equivalent to a binary 1.

Master/slave Bus access method whereby the right to transmit is assigned to one device only, the **master,** and all the other devices, the **slaves** may only transmit when requested.

MDF Main distribution frame

MIPS Million instructions per second

MMI Man-machine-interface

MMS Manufacturing message services – a protocol entity forming part of the application layer. It is intended for use specifically in the manufacturing or process control industry. It enables a supervisory computer to control the operation of a distributed community of computer-based devices.

MNP Microcom networking protocol

Modem eliminator A device used to connect a local terminal and a computer port in lieu of the pair of modems to which they would ordinarily connect, allow DTE to DTE data and control signal connections otherwise not easily achieved by standard cables or connections.

Modem MODulator/DEModulator – a device used to convert serial digital data from a transmitting terminal to a signal suitable for transmission over a telephone channel or to reconvert the transmitted signal to serial digital data for the receiving terminal.

MOS Metal oxide semiconductor

MOV Metal oxide varistor

MSB Most significant bits – the digits on the left hand side of the written HEX or BIN codes.

MSD Most significant digit

MTBF Mean time between failures

MTTR Mean time to repair

Multidrop A single communication line or bus used to connect three or more points.

Multiplexer (MUX) A device used for division of a communication link into two or more channels either by using frequency division or time division.

N

NAK Negative acknowledge (ASCII Control-U)

Network architecture A set of design principles including the organization of functions and the description of data formats and procedures used as the basis for the design and implementation of a network (ISO).

Network layer Layer 3 in the ISO/OSI reference model, the logical network entity that services the transport layer responsible for ensuring that data passed to it from the transport layer is routed and delivered throughout the network.

Network topology The physical and logical relationship of nodes in a network; the schematic arrangement of the links and nodes of a network typically in the form of a star, ring, tree or bus topology.

Network An interconnected group of nodes or stations.

NMRR Normal mode rejection ratio

Node A point of interconnection to a network.

Noise A name given to the extraneous electrical signals that may be generated or picked up in a transmission line. If the noise signal is large compared with the data carrying signal, the latter may be corrupted resulting in transmission errors.

NOS Network operating system

NRM Unbalanced normal response mode

NRZ Non return to zero – pulses in alternating directions for successive 1 bits but no change from existing signal voltage for 0 bits.

NRZI Non return to zero inverted

Null modem A device that connects two DTE devices directly by emulating the physical connections of a DCE device.

Nyquist sampling theorem In order to recover all the information about a specified signal it must be sampled at least at twice the maximum frequency component of the specified signal.

O

OD Object dictionary

Ohm (Ω) Unit of resistance such that a constant current of one ampere produces a potential difference of one Volt across a conductor.

Optical isolation Two networks with no electrical continuity in their connection because an optoelectronic transmitter and receiver have been used.

OSI Open systems interconnection

P

Packet A group of bits (including data and call control signals) transmitted as a whole on a packet switching network. Usually smaller than a transmission block.

PAD Packet access device – an interface between a terminal or computer and a packet switching network.

Parallel transmission The transmission model where a number of bits is sent simultaneously over separate parallel lines. Usually unidirectional such as the Centronics interface for a printer.

Parity bit A bit that is set to a '0' or '1' to ensure that the total number of 1 bits in the data field is even or odd.

Parity check The addition of non-information bits that make up a transmission block to ensure that the total number of bits is always even (even parity) or odd (odd parity). Used to detect transmission errors but rapidly losing popularity because of its weakness in detecting errors.

Passive filter A circuit using only passive electronic components such as resistors, capacitors and inductors.

PBX Private branch exchange

PCIP Personal computer instrument products

PDU Protocol data unit

Peripherals The input/output and data storage devices attached to a computer e.g. disk drives, printers, keyboards, display, communication boards, etc.

Phase modulation The sine wave or carrier changes phase in accordance with the information to be transmitted.

Phase shift keying A modulation technique (also referred to as PSK) used to convert binary data into an analog form comprising a single sinusoidal frequency signal whose phase varies according to the data being transmitted.

Physical layer Layer 1 of the ISO/OSI reference model, concerned with the electrical and mechanical specifications of the network termination equipment.

PID Proportional integral derivative – a form of closed loop control.

PLC Programmable logic controller

Point to point A connection between only two items of equipment.

Polling A means of controlling devices on a multipoint line. A controller queries devices for a response.

Polyethylene A family of insulators derived from the polymerization of ethylene gas and characterized by outstanding electrical properties, including high IR, low dielectric constant, and low dielectric loss across the frequency spectrum.

Polyvinyl chloride (PVC) A general-purpose family of insulations whose basic constituent is polyvinyl chloride or its copolymer with vinyl acetate. Plasticizers, stabilisers, pigments and fillers are

	added to improve mechanical and/or electrical properties of this material.
Port	A place of access to a device or network, used for input/output of digital and analog signals.
Presentation layer	Layer 6 of the ISO/OSI reference model, concerned with negotiating suitable transfer syntax for use during an application. If this is different from the local syntax, the translation to/from this syntax.
Profibus	Process field bus developed by a consortium of mainly German companies with the aim of standardization.
Protocol entity	The code that controls the operation of a protocol layer.
Protocol	A formal set of conventions governing the formatting, control procedures and relative timing of message exchange between two communicating systems.
PSDN	Public switched data network Any switching data communications system, such as telex and public telephone networks, which provides circuit switching to many customers.
PSK	See Phase shift keying
PSTN	Public switched telephone network – this is the term used to describe the (analog) public telephone network.
PTT	Post, Telephone and Telecommunications Authority or: push to talk signal
PV	Primary variable

Q

QAM	Quadrature amplitude modulation
QPSK	Quadrature phase shift keying

R

R/W	Read/write
RAM	Random access memory – semiconductor read/write volatile memory. Data is lost if the power is turned off.
Reactance	The opposition offered to the flow of alternating current by inductance or capacitance of a component or circuit.
REJ	Reject
Repeater	An amplifier that regenerates the signal and thus expands the network.
Resistance	The ratio of voltage to electrical current for a given circuit measured in Ohms.
Response time	The elapsed time between the generation of the last character of a message at a terminal and the receipt of the first character of the reply. It includes terminal delay and network delay.

RF Radio frequency

RFI Radio frequency interference

Ring Network topology commonly used for interconnection of communities of digital devices distributed over a localized area, e.g. a factory or office block. Each device is connected to its nearest neighbors until all the devices are connected in a closed loop or ring. Data is transmitted in one direction only. As each message circulates around the ring, it is read by each device connected in the ring.

RMS Root mean square

RNR Receiver not ready

ROM Read only memory – computer memory in which data can be routinely read but written to only once using special means when the ROM is manufactured. A ROM is used for storing data or programs on a permanent basis.

Router A linking device between network segments which may differ in layers 1, 2a and 2b of the ISO/OSI reference model.

RR Receiver ready

RS Recommended standard (e.g. RS-232C) – newer designations use the prefix EIA (e.g. EIA-RS-232C or just EIA-232C).

RS-232-C Interface between DTE and DCE, employing serial binary data exchange. Typical maximum specifications are 15 m (50 feet) at 19200 Baud.

RS-422 Interface between DTE and DCE employing the electrical characteristics of balanced voltage interface circuits.

RS-423 Interface between DTE and DCE, employing the electrical characteristics of unbalanced voltage digital interface circuits.

RS-449 General purpose 37 pin and 9 pin interface for DCE and DTE employing serial binary interchange.

RS-485 The recommended standard of the EIA that specifies the electrical characteristics of drivers and receivers for use in balanced digital multipoint systems.

RSSI Receiver signal strength indicator

RTS Request to send

RTU Remote terminal unit – terminal unit situated remotely from the main control system.

RxRDY Receiver ready

S

S/N Signal to noise (ratio)

SAA Standards Association of Australia

SAP Service access point

SDLC	Synchronous data link control – IBM standard protocol superseding the bisynchronous standard.
SDM	Space division multiplexing
SDS	Smart distributed system
Serial transmission	The most common transmission mode in which information bits are sent sequentially on a single data channel.
Session layer	Layer 5 of the ISO/OSI reference model, concerned with the establishment of a logical connection between two application entities and with controlling the dialogue (message exchange) between them.
SFD	The start of frame delimiter
Short haul modem	A signal converter that conditions a digital signal for transmission over dc continuous private line metallic circuits, without interfering with adjacent pairs of wires in the same telephone cables.
Signal to noise ratio	The ratio of signal strength to the level of noise.
Simplex transmissions	Data transmission in one direction only.
Slew rate	This is defined as the rate at which the voltage changes from one value to another.
SNA	Subnetwork access, or systems network architecture
SNDC	Subnetwork dependent convergence
SNIC	Subnetwork independent convergence
SNR	Signal to noise ratio
Software	Refers to the programs that are written by a user to control the actions of a microprocessor or a computer. These may be written in one of many different programming languages and may be changed by the user from time to time.
SOH	Start of header (ASCII Control-A)
Space	Absence of signal. This is equivalent to a binary 0.
Spark test	A test designed to locate imperfections (usually pinholes) in the insulation of a wire or cable by application of a voltage for a very short period of time while the wire is being drawn through the electrode field.
SRC	Source node of a message
SREJ	Selective reject
Star	A type of network topology in which there is a central node that performs all switching (and hence routing) functions.
Statistical multiplexer	A device used to enable a number of lower bit rate devices, normally situated in the same location, to share a single,

	higher bit rate transmission line. The devices usually have human operators and hence data is transmitted on the shared line on a statistical basis rather than, as is the case with a basic multiplexer, on a pre-allocated basis. It endeavors to exploit the fact that each device operates at a much lower mean rate than its maximum rate.
STP	Shielded twisted pair
Straight through pinning	RS-232 and RS-422 configuration that match DTE to DCE, pin for pin (pin 1 with pin 1, pin 2 with pin 2, etc.).
STX	Start of text (ASCII Control-B).
Switched line	A communication link for which the physical path may vary with each usage, such as the public telephone network.
SYN	Synchronous Idle
Synchronization	The coordination of the activities of several circuit elements.
Synchronous transmission	Transmission in which data bits are sent at a fixed rate, with the transmitter and receiver synchronized. Synchronized transmission eliminates the need for start and stop bits.

T

Talker	A device on the GPIB bus that simply sends information on to the bus without actually controlling the bus.
TCP	Transmission control protocol
TCU	Trunk coupling unit
TDM	Time division multiplexer A device that accepts multiple channels on a single transmission line by connecting terminals, one at a time, at regular intervals, interleaving bits (bit TDM) or characters (character TDM) from each terminal.
Telegram	In general a data block which is transmitted on the network. Usually comprises address, information and check characters.
Temperature rating	the maximum and minimum temperature at which an insulating material may be used in continuous operation without loss of its basic properties.
TIA	Telecommunications Industry Association
Time sharing	A method of computer operation that allows several interactive terminals to use one computer.
TNS	Transaction bytes
Token ring	Collision free, deterministic bus access method as per IEEE 802.2 ring topology.

TOP Technical Office Protocol – a user association in USA which is primarily concerned with open communications in offices.

Topology Physical configuration of network nodes, e.g. bus, ring, star, tree.

Transceiver Transmitter/receiver – network access point for IEEE 803.2 networks.

Transient An abrupt change in voltage of short duration.

Transport layer Layer 4 of the ISO/OSI reference model, concerned with providing a network independent reliable message interchange service to the application oriented layers (Layers 5 through 7).

Trunk A single circuit between two points, both of which are switching centers or individual distribution points. A trunk usually handles many channels simultaneously.

TTL Transistor-transistor logic

Twisted pair A data transmission medium, consisting of two insulated copper wires twisted together. This improves its immunity to interference from nearby electrical sources that may corrupt the transmitted signal.

U

UART Universal asynchronous receiver/transmitter – an electronic circuit that translates the data format between a parallel representation, within a computer, and the serial method of transmitting data over a communications line.

UHF Ultra high frequency

Unbalanced circuit A transmission line in which voltages on the two conductors are unequal with respect to ground e.g. a coaxial cable.

Unloaded line A line with no loaded coils that reduce line loss at audio frequencies.

UP Unnumbered poll

USB Universal serial bus

USRT Universal synchronous receiver/transmitter. See UART.

UTP Unshielded twisted pair

V

V.35 ITU standard governing the transmission at 48 kbps over 60 to 108 kHz group band circuits.

Velocity of propagation The speed of an electrical signal down a length of cable compared to speed in free space expressed as a percentage.

VFD Virtual field device – a software image of a field device describing the objects supplied by it e.g. measured data,

	events, status etc. which can be accessed by another network.
VHF	Very high frequency
VLAN	Virtual LAN
Volatile memory	An electronic storage medium that loses all data when power is removed.
Voltage rating	The highest voltage that may be continuously applied to a wire in conformance with standards of specifications.
VRC	Vertical redundancy check
VSD	Variable speed drive
VT	Virtual terminal

W

WAN	Wide area network
Word	The standard number of bits that a processor or memory manipulates at one time. Typically, a word has 16 bits.

X

X.21	ITU standard governing interface between DTE and DCE devices for synchronous operation on public data networks.
X.25	ITU standard governing interface between DTE and DCE device for terminals operating in the packet mode on public data networks.
X.25 Pad	A device that permits communication between non X.25 devices and the devices in an X.25 network.
X.3/X.28/X.29	A set of internationally agreed standard protocols defined to allow a character oriented device, such as a visual display terminal, to be connected to a packet switched data network.
X-ON/X-OFF	Transmitter on/transmitter off – control characters used for flow control, instructing a terminal to start transmission (X-ON or control-S) and end transmission (X-OFF or control-Q).
XOR	Exclusive-OR